

# PMA Note

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## 1 The Real and Complex Number Systems

### 1.1 Introduction

**Example 1.1.** It is well known that there is no rational  $p$  such that  $p^2 = 2$ .

Define  $A := \{p \in \mathbb{Q}^+ \mid p^2 < 2\}$  and  $B := \{p \in \mathbb{Q}^+ \mid p^2 > 2\}$ . Then  $A$  contains no largest number and  $B$  contains no smallest.

*Proof.* It suffices to show that

$$\forall p \in A, \exists q \in A \text{ s.t. } p < q$$

and

$$\forall p \in B, \exists q \in B \text{ s.t. } q < p.$$

For every  $p \in \mathbb{Q}^+$ , take

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}. \quad (1)$$

Then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}. \quad (2)$$

Hence,

$$\begin{aligned} p \in A &\implies p^2 - 2 < 0 \\ &\implies p < q \text{ and } q^2 < 2 & (\because (1), (2)) \\ &\implies q \in A \end{aligned}$$

and

$$\begin{aligned} p \in B &\implies p^2 - 2 > 0 \\ &\implies 0 < q < p \text{ and } q^2 > 2 & (\because (1), (2)) \\ &\implies q \in B. \end{aligned}$$

We have shown the desired result.  $\square$

## 1.2 Ordered Sets

**Definition 1.2.** Let  $S$  be a set. An *order* on  $S$  is a relation, denoted by  $<$ , with the following two properties:

- (i) If  $x \in S$  and  $y \in S$  then one and only one of the statements

$$x < y, \quad x = y, \quad y < x$$

is true.

- (ii) If  $x, y, z \in S$ , if  $x < y$  and  $y < z$ , then  $x < z$ .

**Definition 1.3.** An *ordered set* is a set  $S$  in which an order is defined.

**Definition 1.4.** Suppose  $S$  is an ordered set, and  $E \subset S$ . If there exists a  $\beta \in S$  such that  $x \leq \beta$  for every  $x \in E$ , we say that  $E$  is *bounded above*, and call  $\beta$  an *upper bound* of  $E$ .

Lower bounds are defined in the same way (with  $\geq$  in place of  $\leq$ ).

**Definition 1.5.** Suppose  $S$  is an ordered set,  $E \subset S$ , and  $E$  is bounded above. Suppose there exists an  $\alpha \in S$  with the following properties:

- (i)  $\alpha$  is an upper bound of  $E$ .  
(ii) If  $\gamma < \alpha$  then  $\gamma$  is not an upper bound of  $E$ .

Then  $\alpha$  is called the *least upper bound* of  $E$  [that there is at most one such  $\alpha$  is clear from (ii)] or the *supremum* of  $E$ , and we write

$$\alpha = \sup E.$$

The *greatest lower bound*, or *infimum*, of a set  $E$  which is bounded below is defined in the same manner: The statement

$$\alpha = \inf E$$

means that  $\alpha$  is a lower bound of  $E$  and that no  $\beta$  with  $\beta > \alpha$  is a lower bound of  $E$ .

**Definition 1.6.** An ordered set  $S$  is said to have the *least-upper-bound property* if the following is true:

If  $E \subset S$ ,  $E$  is not empty, and  $E$  is bounded above, then  $\sup E$  exists in  $S$ .

*Remark 1.7.* The ordered set  $\mathbb{Q}$  does not have the least-upper-bound property.

The following theorem shows that every ordered set with the least-upper-bound property also has the greatest-lower-bound property.

**Theorem 1.8.** Suppose  $S$  is an ordered set with the least-upper-bound property,  $B \subset S$ ,  $B$  is not empty, and  $B$  is bounded below. Let  $L$  be the set of all lower bounds of  $B$ . Then

$$\alpha = \sup L$$

exists in  $S$ , and  $\alpha = \inf B$ .

In particular,  $\inf B$  exists in  $S$ .

*Proof.* Since  $B$  is bounded below,  $L$  is nonempty. Let  $x \in B$ . Then  $y \leq x$  for every  $y \in L$ . Thus, every  $x \in B$  is an upper bound of  $L$ . Since  $L$  is a nonempty subset of  $S$  that is bounded above, by hypothesis,

$$\alpha = \sup L$$

exists in  $S$ .

If  $\gamma < \alpha$  then  $\gamma$  is not an upper bound of  $L$ , hence  $\gamma \notin B$ . It follows that  $\alpha \leq x$  for every  $x \in B$ . Thus,  $\alpha \in L$ .

If  $\alpha < \beta$  then  $\beta \notin L$ , since  $\alpha$  is an upper bound of  $L$ .

We conclude that  $\alpha$  is a lower bound of  $B$  (since  $\alpha \in L$ ) and if  $\alpha < \beta$  then  $\beta$  is not a lower bound of  $B$  (since  $\beta \notin L$ ), which implies  $\alpha = \inf B$ .  $\square$

*Remark 1.9.* Fix  $x \in B$ . Then  $y \leq x$  for all  $y \in L$ . Thus,  $\sup L \leq x$ . Since the choice of  $x \in B$  was arbitrary, we have

$$\sup L = \alpha \leq x \text{ for every } x \in B,$$

which implies  $\alpha$  is a lower bound of  $B$ . Hence,  $\alpha \in L$ .

### 1.3 Fields

**Definition 1.10.** A *field* is a set  $F$  with two operations, called *addition* and *multiplication*, which satisfy the following so-called "field axioms" (A), (M), and (D):

(A) Axioms for addition

- (A1) If  $x \in F$  and  $y \in F$ , then their sum  $x + y$  is in  $F$ .
- (A2) Addition is commutative:  $x + y = y + x$  for all  $x, y \in F$ .
- (A3) Addition is associative:  $(x + y) + z = x + (y + z)$  for all  $x, y, z \in F$ .
- (A4)  $F$  contains an element  $0$  such that  $0 + x = x$  for every  $x \in F$ .
- (A5) To every  $x \in F$  corresponds an element  $-x \in F$  such that

$$x + (-x) = 0.$$

(M) Axioms for multiplication

- (M1) If  $x \in F$  and  $y \in F$ , then their product  $xy$  is in  $F$ .
- (M2) Multiplication is commutative:  $xy = yx$  for all  $x, y \in F$ .
- (M3) Multiplication is associative:  $(xy)z = x(yz)$  for all  $x, y, z \in F$ .
- (M4)  $F$  contains an element  $1 \neq 0$  such that  $1x = x$  for every  $x \in F$ .
- (M5) If  $x \in F$  and  $x \neq 0$  then there exists an element  $1/x \in F$  such that

$$x \cdot (1/x) = 1.$$

(D) The distributive law

$$x(y + z) = xy + xz$$

holds for all  $x, y, z \in F$ .

**Definition 1.11.** An *ordered field* is a *field*  $F$  which is also an *ordered set*, such that

- (i)  $x + y < x + z$  if  $x, y, z \in F$  and  $y < z$ .
- (ii)  $xy > 0$  if  $x \in F$ ,  $y \in F$ ,  $x > 0$ , and  $y > 0$ .

## 1.4 The Real Field

**Theorem 1.12.** *There exists an ordered field  $\mathbb{R}$  which has the least-upper-bound property. Moreover,  $\mathbb{R}$  contains  $\mathbb{Q}$  as a subfield.*

**Theorem 1.13.**

(a) *If  $x \in \mathbb{R}$ ,  $y \in \mathbb{R}$ , and  $x > 0$ , then there is a positive integer  $n$  such that*

$$nx > y.$$

(b) *If  $x \in \mathbb{R}$ ,  $y \in \mathbb{R}$ , and  $x < y$ , then there exists a  $p \in \mathbb{Q}$  such that  $x < p < y$ .*

*Proof.*

(a) Let  $x, y \in \mathbb{R}$  and  $x > 0$ . Suppose to the contrary that  $nx \leq y$  for each  $n \in \mathbb{N}$  and define

$$A := \{nx \mid n \in \mathbb{N}\}.$$

Then  $A$  is nonempty and bounded above by  $y$ . Thus, the supremum of  $A$  exists in  $\mathbb{R}$ . Put  $\alpha = \sup A$ .

Since  $x > 0$ , we have  $\alpha - x < \alpha$ , hence,  $\alpha - x$  is not an upper bound of  $A$ . Therefore,  $\alpha - x < mx$  for some  $m \in \mathbb{N}$ . Then  $\alpha < (m+1)x \in A$ , which is a contradiction, since  $\alpha$  is an upper bound of  $A$ .

(b) Since Rudin's proof of part (b) is already explicit, we omit it here. □

**Theorem 1.14.** *For every real  $x > 0$  and every integer  $n > 0$  there is one and only one positive real  $y$  such that  $y^n = x$ .*

This number  $y$  is written  $\sqrt[n]{x}$  or  $x^{1/n}$ .

*Proof.* We omit the proof here. □

**Corollary.** *If  $a$  and  $b$  are positive and real numbers and  $n$  is a positive integer, then*

$$(ab)^{\frac{1}{n}} = a^{\frac{1}{n}} b^{\frac{1}{n}}.$$

*Proof.* We omit the proof here. □

**Lemma 1.15.** <sup>1</sup>

(a) *If  $E \subset \mathbb{Z}$  has the infimum, then  $\inf E \in E$ .*

(b) *If  $x \in \mathbb{R}$  then there exists an  $n \in \mathbb{Z}$  such that*

$$n \leq x < n+1.$$

*Proof.*

(a) Let  $\alpha = \inf E$ . Suppose to the contrary that  $\alpha \notin E$ . Since  $\alpha + 1$  is not a lower bound of  $E$ , choose an  $x_0 \in E$  such that

$$\alpha < x_0 < \alpha + 1. \tag{3}$$

Since  $x_0$  is not a lower bound of  $E$ , choose an  $x_1 \in E$  again such that

$$\alpha < x_1 < x_0. \tag{4}$$

Subtracting  $x_0$  from (4), we obtain  $\alpha - x_0 < x_1 - x_0 < 0$ . Since  $-1 < \alpha - x_0$  from (3), it follows that  $-1 < x_1 - x_0 < 0$ . This contradicts the fact that  $x_1 - x_0 \in \mathbb{Z}$ . We conclude that  $\alpha \in E$ .

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<sup>1</sup>This lemma has been added for convenience and does not appear in the original text.

(b) Let  $x \in \mathbb{R}$  and define  $E := \{n \in \mathbb{Z} \mid n > x\}$ . By the Archimedean Property,  $E$  is nonempty. Since  $E$  is bounded below by  $x$ , the infimum of  $E$  exists in  $\mathbb{R}$ . Since  $E \subset \mathbb{Z}$ ,  $\inf E$  is the least element of  $E$  by (a). Thus, we have

$$\inf E - 1 \leq x < \inf E.$$

Putting  $n = \inf E - 1 \in \mathbb{Z}$ , we obtain the desired inequality.  $\square$

*Remark 1.16 (Decimals).* Let  $x > 0$  be real. Let  $n_0$  be the largest integer such that  $n_0 \leq x$ , by lemma 1.15. Having chosen  $n_0, n_1, \dots, n_{k-1}$ , let  $n_k$  be the largest integer such that

$$n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k} \leq x.$$

Let  $E$  be the set of these numbers

$$n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k} \quad (k = 0, 1, 2, \dots). \quad (5)$$

Notice that  $x$  is an upper bound of  $E$  and if  $k \in \mathbb{N}$  then

$$n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k} \leq x < n_0 + \frac{n_1}{10} + \dots + \frac{n_k + 1}{10^k}.$$

Hence, we have

$$x - n_0 - \frac{n_1}{10} - \dots - \frac{n_k}{10^k} < \frac{1}{10^k} \text{ for } k \in \mathbb{N}. \quad (6)$$

Suppose  $y < x$ . Then by the Archimedean Property, there exists a positive integer  $k$  such that  $1/10^k < x - y$ . From (6), we obtain

$$x - n_0 - \frac{n_1}{10} - \dots - \frac{n_k}{10^k} < x - y,$$

hence

$$y < n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k}.$$

Therefore,  $x = \sup E$ . The decimal expansion of  $x$  is

$$n_0 \cdot n_1 n_2 n_3 \dots \quad (7)$$

Conversely, for any infinite decimal (7), the set  $E$  of numbers (5) is bounded above, and (7) is the decimal expansion of  $\sup E$ .

## 1.5 Exercises

**Exercise 1.** Let  $A$  be a nonempty set of real numbers which is bounded below. Let  $-A$  be the set of all numbers  $-x$ , where  $x \in A$ . Prove that

$$\inf A = -\sup(-A).$$

*Proof.* Since  $A$  is bounded below, the infimum of  $A$  exists in  $\mathbb{R}$ . If  $a \in A$  then  $\inf A \leq a$ . Hence,

$$-a \leq -\inf A \text{ for all } a \in A, \quad (8)$$

and it follows that  $-A$  is bounded above. Thus,  $\sup(-A)$  exists in  $\mathbb{R}$ . Since  $-\inf A$  is an upper bound of  $-A$  from (8), we have  $\sup(-A) \leq -\inf A$ , hence

$$\inf A \leq -\sup(-A). \quad (9)$$

Since  $-a \leq \sup(-A)$  for every  $a \in A$ , we obtain  $-\sup(-A) \leq a$  for all  $a \in A$ . Therefore,  $-\sup(-A)$  is a lower bound of  $A$  and

$$-\sup(-A) \leq \inf A. \quad (10)$$

Combining (9) and (10), we conclude that  $\inf A = -\sup(-A)$ .  $\square$

**Exercise 2.** Fix  $b > 1$ .

(a) If  $m, n, p, q$  are integers,  $n > 0, q > 0$ , and  $r = m/n = p/q$ , prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Hence it makes sense to define  $b^r = (b^m)^{1/n}$ .

(b) Prove that  $b^{r+s} = b^r b^s$  if  $r$  and  $s$  are rational.

(c) If  $x$  is real, define  $B(x)$  to be the set of all numbers  $b^t$ , where  $t$  is rational and  $t \leq x$ . Prove that

$$b^r = \sup B(r)$$

when  $r$  is rational. Hence it makes sense to define

$$b^x = \sup B(x)$$

for every real  $x$ .

(d) Prove that  $b^{x+y} = b^x b^y$  for all real  $x$  and  $y$ .

*Proof.*

(a) Let  $b^{mq} = b^{np} = x$ . Since  $n$  and  $q$  are positive integers and  $b > 0$ , by Theorem 1.14, we have

$$b^m = x^{1/q} \text{ and } b^p = x^{1/n}.$$

It follows that

$$(b^m)^{1/n} = (x^{1/q})^{1/n} = (x^{1/n})^{1/q} = (b^p)^{1/q},$$

since  $[(x^{1/q})^{1/n}]^{nq} = [(x^{1/n})^{1/q}]^{nq} = x$ .

(b) Let  $m, n, p, q$  be integers and  $n > 0, q > 0$  and let  $r = m/n$  and  $s = p/q$ . Then by the Corollary to Theorem 1.14, we obtain

$$\begin{aligned} b^{r+s} &= b^{\frac{m}{n} + \frac{p}{q}} = b^{\frac{mq+np}{nq}} \\ &= (b^{mq+np})^{\frac{1}{nq}} \\ &= (b^{mq} b^{np})^{\frac{1}{nq}} \\ &= (b^{mq})^{\frac{1}{nq}} (b^{np})^{\frac{1}{nq}} \\ &= b^{\frac{mq}{nq}} b^{\frac{np}{nq}} = b^{\frac{m}{n}} b^{\frac{p}{q}} = b^r b^s. \end{aligned}$$

(c) Note that by Theorem 1.14,  $b^r > 0$  if  $b > 1$  and  $r$  is rational.

Let  $n$  be a positive integer and let  $0 < a < 1$  be real. We first show that  $0 < a^{1/n} < 1$ .

Suppose to the contrary that  $a^{1/n} \geq 1$ . Then we obtain

$$1 \leq (a^{1/n})^n = a < 1,$$

a contradiction. Therefore, if  $0 < a < 1$  then  $0 < a^{1/n} < 1$ , where  $n$  is a positive integer.

Suppose  $r$  and  $s$  are rational. Let  $m, n, p$ , and  $q$  be integers and let  $n$  and  $q$  be positive. Then we write  $r = m/n, s = p/q$ .

Suppose  $r < s$ . Then

$$r - s = \frac{m}{n} - \frac{p}{q} = \frac{mq - np}{nq} < 0,$$

hence

$$0 < b^{mq-np} = \frac{1}{b^{np-mq}} < 1$$

and

$$0 < b^{r-s} = b^{\frac{mq-np}{nq}} = (b^{mq-np})^{1/nq} < 1.$$

Thus, if  $r < s$  then

$$b^r = b^{(r-s)+s} = b^{r-s}b^s < b^s.$$

If  $t$  is rational and  $t \leq r$  then  $b^t \leq b^r$ . It follows that  $B(r)$  is bounded above by  $b^r$  and  $\sup B(r)$  exists in  $\mathbb{R}$ . Since  $b^r$  is an upper bound of  $B(r)$ ,

$$\sup B(r) \leq b^r.$$

Since  $b^r \in B(r)$ ,

$$b^r \leq \sup B(r).$$

We conclude that  $b^r = \sup B(r)$ .

(d) Let  $t_1$  and  $t_2$  be rational. If  $t_1 \leq x$  and  $t_2 \leq y$  then  $t_1 + t_2 \leq x + y$ . Thus, we have

$$b^{t_1}b^{t_2} = b^{t_1+t_2} \leq \sup B(x+y) = b^{x+y}.$$

Fix  $t_1 \leq x$ . Then

$$b^{t_2} \leq b^{x+y}/b^{t_1} \text{ for all } t_2 \leq y.$$

Hence,

$$b^y = \sup B(y) \leq b^{x+y}/b^{t_1}.$$

Since the choice of  $t_1$  was arbitrary,

$$b^{t_1} \leq b^{x+y}/b^y \text{ for all } t_1 \leq x,$$

and it follows that

$$b^x = \sup B(x) \leq b^{x+y}/b^y.$$

Therefore, we obtain

$$b^x b^y \leq b^{x+y}. \tag{11}$$

Notice that

$$0 < b^{t_1} \leq b^x \text{ for every } t_1 \leq x$$

and

$$0 < b^{t_2} \leq b^y \text{ for every } t_2 \leq y.$$

Hence, for every  $t_1 \leq x$  and  $t_2 \leq y$ ,

$$b^{t_1+t_2} = b^{t_1}b^{t_2} \leq b^x b^y.$$

Now, to obtain

$$b^{x+y} = \sup B(x+y) \leq b^x b^y \tag{12}$$

from above inequality, we need the following statement:

$$b^x = \sup \{b^t \mid t \in \mathbb{Q}, t < x\}.$$

Let us prove it. Notice that

$$b^n - 1 = (b-1)(b^{n-1} + \cdots + 1) \geq n(b-1)$$

for any positive integer  $n$ . Put  $c = b^{1/n} > 1$ . Then we have

$$b - 1 \geq n(b^{1/n} - 1).$$

Hence, if  $t > 1$  and  $n > (b-1)/(t-1)$  then

$$\frac{n(b^{1/n} - 1)}{t - 1} \leq \frac{b - 1}{t - 1} < n,$$

and it follows that

$$b^{1/n} < t.$$

Note that if  $t > 1$  then  $b^{1/n} < t$  for sufficiently large  $n$ .

Let  $B_0(x) = \{b^t \mid t \in \mathbb{Q}, t < x\}$ . We need to show that  $b^x$  is the supremum of  $B_0(x)$ . It is clear that  $b^x$  is an upper bound of  $B_0(x)$ . Thus, it remains to prove that if  $y < b^x$  then  $y$  is not an upper bound.

Suppose  $0 < y < b^x$ . Then  $b^x/y > 1$  and it follows that

$$b^{1/n} < b^x/y$$

for sufficiently large  $n$ . Hence, we have

$$y < \frac{b^x}{b^{1/n}} < b^x.$$

By the density of rationals, choose a rational  $t$  such that  $x - (1/n) < t < x$ . Since  $x < s < t + (1/n)$  for some rational  $s$ ,

$$b^x \leq b^s < b^{t+(1/n)} = b^t b^{1/n}$$

and

$$y < \frac{b^x}{b^{1/n}} < b^t.$$

Thus,  $y$  is not an upper bound of  $B_0(x)$  and therefore  $b^x = \sup B_0(x)$ .

Now, let  $t$  be a rational and let  $t < x + y$ . Pick  $t_2 \in \mathbb{Q}$  such that  $t - x < t_2 < y$  and set  $t_1 = t - t_2$ . Then for every  $t \in \mathbb{Q}$  with  $t < x + y$ , there are rational  $t_1 < x$  and  $t_2 < y$  such that  $t = t_1 + t_2 < x + y$ , hence  $b^t = b^{t_1+t_2} = b^{t_1}b^{t_2} \leq b^x b^y$ . Thus, we obtain (12).

Combining (11) and (12), we conclude that

$$b^{x+y} = b^x b^y.$$

□

**Exercise 3.** Prove that no order can be defined in the complex field that turns it into an ordered field.

*Proof.* Suppose to the contrary that there exists an order  $<$  on  $\mathbb{C}$  which turns  $\mathbb{C}$  into an ordered field. Let  $\mathbf{0}$  and  $\mathbf{1}$  denote the additive and multiplicative identities of the complex field, respectively. Then  $\mathbf{0} = (0, 0)$  and  $\mathbf{1} = (1, 0)$ .

Note that  $i = (0, 1) \neq \mathbf{0}$ . If  $x \neq \mathbf{0}$  then  $x^2 > \mathbf{0}$ . It follows that

$$i^2 = (0, 1)(0, 1) = (-1, 0) > \mathbf{0}.$$

Since  $(-1, 0) + (1, 0) = \mathbf{0}$ ,  $(-1, 0)$  is the additive inverse of  $(1, 0)$ . Since  $\mathbf{1} > \mathbf{0}$  and if  $x > \mathbf{0}$  then  $-x < \mathbf{0}$  in any ordered field, we obtain

$$(-1, 0) = -(1, 0) = -\mathbf{1} < \mathbf{0}.$$

This contradicts the Trichotomy Property.

□



## 2 Basic Topology

### 2.1 Finite, Countable, and Uncountable Sets

We omit this section here.

### 2.2 Metric Spaces

**Definition 2.1.** A set  $X$ , whose elements we shall call *points*, is said to be a *metric space* if with any two points  $p$  and  $q$  of  $X$  there is associated a real number  $d(p, q)$ , called the *distance* from  $p$  to  $q$ , such that

- (a)  $d(p, q) > 0$  if  $p \neq q$ ;  $d(p, p) = 0$ ;
- (b)  $d(p, q) = d(q, p)$ ;
- (c)  $d(p, q) \leq d(p, r) + d(r, q)$ , for any  $r \in X$ .

Any function with these three properties is called a *distance function*, or a *metric*.

**Example 2.2.**  $\mathbb{R}^k$  is a metric space.

**Definition 2.3.** By the *segment*  $(a, b)$  we mean the set of all real numbers  $x$  such that  $a < x < b$ .

By the *interval*  $[a, b]$  we mean the set of all real numbers  $x$  such that  $a \leq x \leq b$ .

Occasionally we shall also encounter “half-open intervals”  $[a, b)$  and  $(a, b]$ ; the first consists of all  $x$  such that  $a \leq x < b$ , the second of all  $x$  such that  $a < x \leq b$ .

If  $a_i < b_i$  for  $i = 1, \dots, k$ , the set of all points  $\mathbf{x} = (x_1, \dots, x_k)$  in  $\mathbb{R}^k$  whose coordinates satisfy the inequalities  $a_i \leq x_i \leq b_i$  ( $1 \leq i \leq k$ ) is called a *k-cell*. Thus a 1-cell is an interval, a 2-cell is a rectangle, etc.

If  $\mathbf{x} \in \mathbb{R}^k$  and  $r > 0$ , the *open* (or *closed*) *ball*  $B$  with center at  $\mathbf{x}$  and radius  $r$  is defined to be the set of all  $\mathbf{y} \in \mathbb{R}^k$  such that  $|\mathbf{y} - \mathbf{x}| < r$  (or  $|\mathbf{y} - \mathbf{x}| \leq r$ ).

We call a set  $E \subset \mathbb{R}^K$  *convex* if

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in E$$

whenever  $\mathbf{x} \in E$ ,  $\mathbf{y} \in E$ , and  $0 < \lambda < 1$ .

**Example 2.4.** Balls and k-cells are convex.

*Proof.* If  $|\mathbf{y} - \mathbf{x}| < r$ ,  $|\mathbf{z} - \mathbf{x}| < r$ , and  $0 < \lambda < 1$ , then

$$\begin{aligned} |\lambda \mathbf{y} + (1 - \lambda) \mathbf{z} - \mathbf{x}| &= |\lambda(\mathbf{y} - \mathbf{x}) + \lambda \mathbf{x} + (1 - \lambda) \mathbf{z} - \mathbf{x}| \\ &= |\lambda(\mathbf{y} - \mathbf{x}) + (1 - \lambda)(\mathbf{z} - \mathbf{x})| \\ &\leq \lambda |\mathbf{y} - \mathbf{x}| + (1 - \lambda) |\mathbf{z} - \mathbf{x}| \\ &< \lambda r + (1 - \lambda) r = r. \end{aligned}$$

Hence, balls are convex.

Let  $A \subset \mathbb{R}^k$  be a k-cell and let  $\mathbf{x}, \mathbf{y} \in A$  and  $0 < \lambda < 1$ . We write  $\mathbf{x} = (x_1, \dots, x_k)$ ,  $\mathbf{y} = (y_1, \dots, y_k)$  with

$$a_i < b_i \text{ and } \mathbf{x}, \mathbf{y} \in [a_i, b_i] \text{ for } i = 1, \dots, k.$$

If  $x, y \in [a, b]$  with  $a < b$  and  $0 < \lambda < 1$  then

$$\lambda a \leq \lambda x \leq \lambda b,$$

and

$$(1 - \lambda)a \leq (1 - \lambda)y \leq (1 - \lambda)b,$$

hence

$$a \leq \lambda x + (1 - \lambda)y \leq b.$$

It follows that each  $[a_i, b_i]$  ( $1 \leq i \leq k$ ) is convex.

Therefore, every  $k$ -cell is convex.  $\square$

**Definition 2.5.** Let  $X$  be a metric space. All points and sets mentioned below are understood to be elements and subsets of  $X$ .

- (a) A *neighborhood* of  $p$  is a set  $N_r(p)$  consisting of all  $q$  such that  $d(p, q) < r$ , for some  $r > 0$ . The number  $r$  is called the *radius* of  $N_r(p)$ .
- (b) A point  $p$  is a *limit point* of the set  $E$  if *every* neighborhood of  $p$  contains a point  $q \neq p$  such that  $q \in E$ .
- (c) If  $p \in E$  and  $p$  is not a limit point of  $E$ , then  $p$  is called an *isolated point* of  $E$ .
- (d)  $E$  is *closed* if every limit point of  $E$  is a point of  $E$ .
- (e) A point  $p$  is an *interior* point of  $E$  if there is a neighborhood  $N$  of  $p$  such that  $N \subset E$ .
- (f)  $E$  is *open* if every point of  $E$  is an interior point of  $E$ .
- (g) The *complement* of  $E$  (denoted by  $E^c$ ) is the set of all points  $p \in X$  such that  $p \notin E$ .
- (h)  $E$  is *perfect* if  $E$  is closed and if every point of  $E$  is a limit point of  $E$ .
- (i)  $E$  is *bounded* if there is a real number  $M$  and a point  $q \in X$  such that  $d(p, q) < M$  for all  $p \in E$ .
- (j)  $E$  is *dense in  $X$*  if every point of  $X$  is a limit point of  $E$ , or a point of  $E$  (or both).

**Theorem 2.6.** Every neighborhood is an open set.

*Proof.* Consider a neighborhood  $E = N_r(p)$ , and let  $q$  be any point of  $E$ . Then choose an  $h > 0$  such that

$$d(p, q) = r - h.$$

If  $s \in N_h(q)$  then

$$d(p, s) \leq d(p, q) + d(q, s) < r - h + h = r,$$

and it follows that  $N_h(q) \subset E$ . Thus there is a neighborhood of  $q$  that is a subset of  $E$ . Therefore, every  $q \in E$  is an interior point of  $E$ . We conclude that  $E$  is an open set.  $\square$

**Theorem 2.7.** If  $p$  is a limit point of a set  $E$ , then every neighborhood of  $p$  contains infinitely many points of  $E$ .

*Proof.* Suppose there is a neighborhood  $N$  of  $p$  which contains only a finite number of points of  $E$ . Let  $q_1, \dots, q_n$  be those points of  $N \cap E$ , which are distinct from  $p$ , and put

$$r = \min_{1 \leq m \leq n} d(p, q_m) > 0$$

Then  $d(p, q_m) \geq r$  for  $m = 1, \dots, n$  and hence all the points  $q_m$  are not in the neighborhood  $N_r(p)$ . Since

$$N_r(p) \subset N$$

and hence

$$N_r(p) \cap E \setminus \{p\} \subset N \cap E \setminus \{p\} = \{q_1, \dots, q_n\} \subset E,$$

$N_r(p)$  contains no element  $q \neq p$  in  $E$  (notice  $N_r(p) \cap E \setminus \{p\}$  is empty). Thus,  $p$  is not a limit point of  $E$ . This is a contradiction.  $\square$

**Corollary.** *A finite point set has no limit points.*

**Theorem 2.8.** *Let  $\{E_\alpha\}$  be a (finite or infinite) collection of sets  $E_\alpha$ . Then*

$$\left(\bigcup_{\alpha} E_{\alpha}\right)^c = \bigcap_{\alpha} (E_{\alpha}^c). \quad (13)$$

*Proof.* We have

$$\begin{aligned} x \in \left(\bigcup_{\alpha} E_{\alpha}\right)^c &\implies x \notin \left(\bigcup_{\alpha} E_{\alpha}\right) \\ &\implies \forall \alpha, x \notin E_{\alpha} & (\because \text{negation of the definition}) \\ &\implies \forall \alpha, x \in E_{\alpha}^c \\ &\implies x \in \bigcap_{\alpha} E_{\alpha}^c. \end{aligned}$$

Hence  $(\bigcup_{\alpha} E_{\alpha})^c \subset \bigcap_{\alpha} (E_{\alpha}^c)$ . Since

$$\begin{aligned} x \in \bigcap_{\alpha} (E_{\alpha}^c) &\implies \forall \alpha, x \in E_{\alpha}^c \\ &\implies \forall \alpha, x \notin E_{\alpha} \\ &\implies x \notin \bigcup_{\alpha} E_{\alpha} & (\because \text{negation of the definition}) \\ &\implies x \in \left(\bigcup_{\alpha} E_{\alpha}\right)^c, \end{aligned}$$

we obtain the reverse. This establishes the theorem.  $\square$

**Theorem 2.9.** *A set  $E$  is open if and only if its complement is closed.*

*Proof.* First, suppose  $E^c$  is closed. Choose  $x \in E$ . Then  $x \notin E^c$ , and  $x$  is not a limit point of  $E^c$ , since every limit point of  $E^c$  is a point of  $E^c$  (contrapositive). Hence there exists a neighborhood  $N$  of  $x$  such that  $E^c \cap N$  is empty, that is,  $N \subset E$ . Thus  $E$  is open.

Next, suppose  $E$  is open. Let  $x$  be a limit point of  $E^c$ . Then every neighborhood of  $x$  contains a point of  $E^c$ , hence there is no neighborhood  $N$  of  $x$  such that  $N \subset E$ . Thus,  $x$  is not an interior point of  $E$  and therefore  $x \notin E$ , since  $E$  is open. It follows that if  $x$  is a limit point of  $E^c$  then  $x \notin E$ , hence  $x \in E^c$ . Therefore,  $E^c$  is closed.  $\square$

**Corollary.** *A set  $F$  is closed if and only if its complement is open.*

**Theorem 2.10.**

- (a) *For any collection  $\{G_{\alpha}\}$  of open sets,  $\bigcup_{\alpha} G_{\alpha}$  is open.*
- (b) *For any collection  $\{F_{\alpha}\}$  of closed sets,  $\bigcap_{\alpha} F_{\alpha}$  is closed.*
- (c) *For any finite collection  $G_1, \dots, G_n$  of open sets,  $\bigcap_{i=1}^n G_i$  is open.*
- (d) *For any finite collection  $F_1, \dots, F_n$  of closed sets,  $\bigcup_{i=1}^n F_i$  is closed.*

*Proof.* Put  $G = \bigcup_{\alpha} G_{\alpha}$ . Then

$$\begin{aligned} x \in G &\implies \exists \alpha \text{ s.t. } x \in G_{\alpha} \\ &\implies x \text{ is an interior point of } G_{\alpha} \\ &\implies \exists r_0 > 0 \text{ s.t. } N_{r_0}(x) \subset G_{\alpha} \\ &\implies \exists r_0 > 0 \text{ s.t. } N_{r_0}(x) \subset G. \end{aligned}$$

Hence, every  $x \in G$  is an interior point of  $G$  and it follows that  $G$  is open.

(b) follows from (a), Theorem 2.8, and Theorem 2.9. We see that

$$\begin{aligned}
 F_\alpha \text{ is closed} &\implies F_\alpha^c \text{ is open.} && (\because \text{Theorem 2.9}) \\
 &\implies \bigcup_\alpha F_\alpha^c \text{ is open.} && (\because \text{(a)}) \\
 &\implies \left( \bigcap_\alpha F_\alpha \right)^c \text{ is open.} && (\because \text{Theorem 2.8}) \\
 &\implies \bigcap_\alpha F_\alpha \text{ is closed.} && (\because \text{Theorem 2.9})
 \end{aligned}$$

□